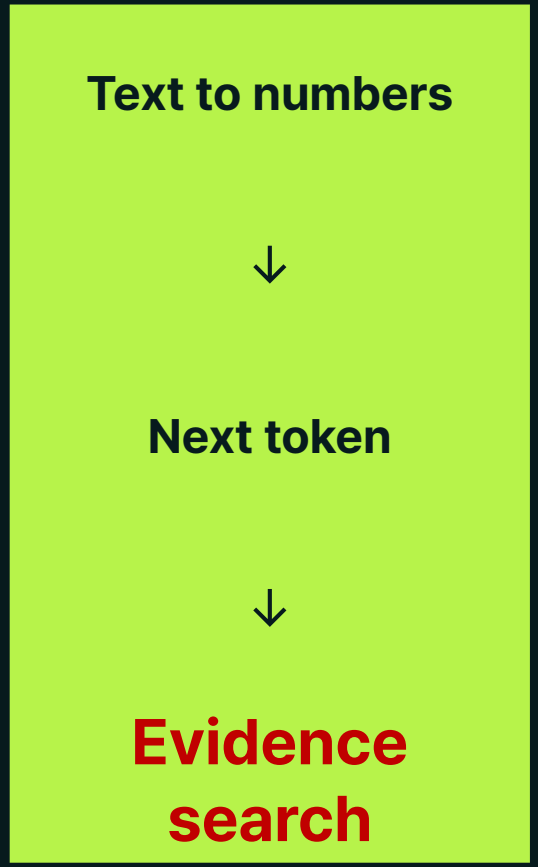
NLP · LLM · RAG
Reading with Evidence

Text to numbers, next word, answers with sources



Financial ML · Week 12

This week runs in four steps

From basics to core concepts, cases, then summary

01	**Basics**	**Vectors, generation** Vectors, generation, search types
02	**Core concepts**	**Tokens to RAG** Embedding, softmax, attention, RAG
03	**Cases**	**Finance text, AI errors** Finance lexicon, RAG errors, fake cases
04	**Summary**	**Checking evidence** Check Q&A

A fluent answer is not a fact check

01

Basics

Basic concepts for understanding language models

A vector is an ordered list of numbers

Embeddings require knowing vector direction and distance

VECTOR

Vector

- Numbers like (1, 0), with direction and length.

DIMENSION

Dimension

- How many numbers. Real embeddings have hundreds or more.

SIMILARITY

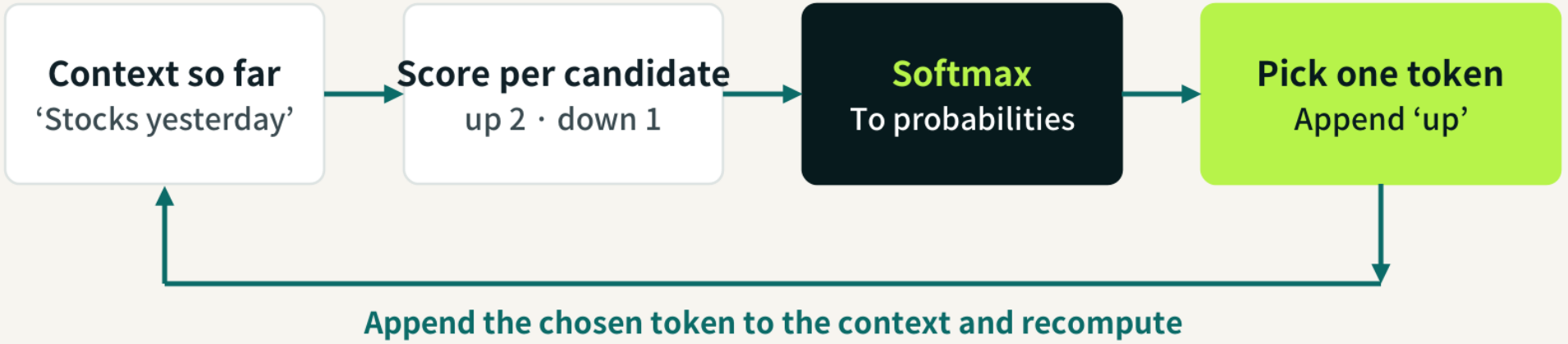
Similarity

- Closer direction means closer meaning.

The figure is 2D, but the idea holds in high dimensions

An LLM picks one token, then recomputes, again and again

Generation chains next-token predictions together



Long answers are built piece by piece

Plausible is not the same as true

Keyword and semantic search have different strengths

How to choose a retrieval method for RAG

KEYWORD · TF-IDF

Keyword search

- Finds exact word matches.
- May miss different wording.

Good for numbers, names

SEMANTIC · EMBEDDING

Semantic search

- Finds similar meaning in other words.
- May miss exact number matches.

Good for paraphrases

Use both and evaluate retrieval separately

02

Tokens to RAG

Source concepts: embedding, similarity, generation, attention, RAG

Computers must turn text into numbers to read it

‘Rate hike’ makes people think of loans and the economy; not computers

NLP

Natural language proc.

- A broad name for all tech that processes human language.

LLM

Large language model

- A big language model trained on lots of text.

RAG

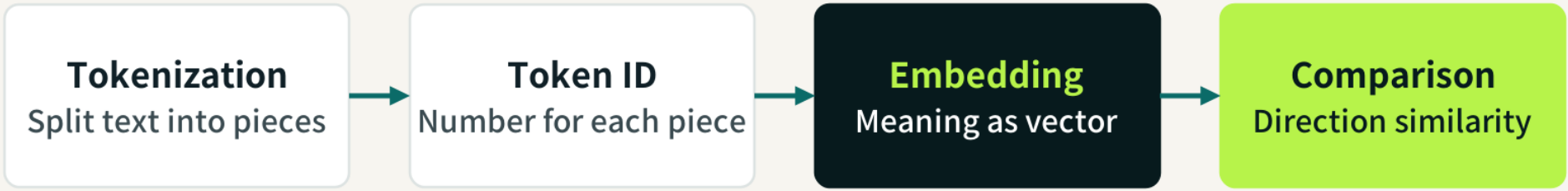
Retrieval-augmented gen.

- A system that fetches external documents before the model answers.

NLP is a field, LLM a model, RAG a system design

Tokens, token IDs and embeddings are distinct steps

Library analogy: call numbers identify books; a topic map groups similar ones



IDs 100 and 101 being close does not mean similar meaning

Similar meanings can point in similar directions

A token is not always one word

Close IDs do not mean close meaning — use embeddings to compare meaning

Cosine similarity compares vector directions

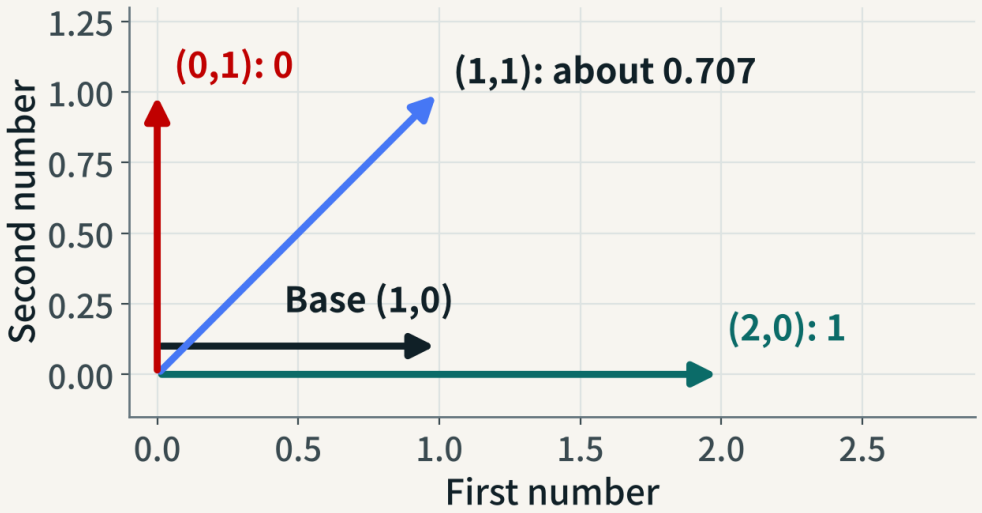
It checks how aligned the directions are, not lengths

Cosine similarity

$$\cos(a, b) = \frac{a \cdot b}{\|a\| \, \|b\|}$$

Caution

- Similarity ≠ probability of truth.
- ‘Increased’ and ‘did not increase’ share words but mean opposites.



※ Same direction gives similarity 1, even if lengths differ

King – man + woman ≈ queen is an intuition from some embeddings, not a law

An LLM computes next-token odds and adds one piece at a time

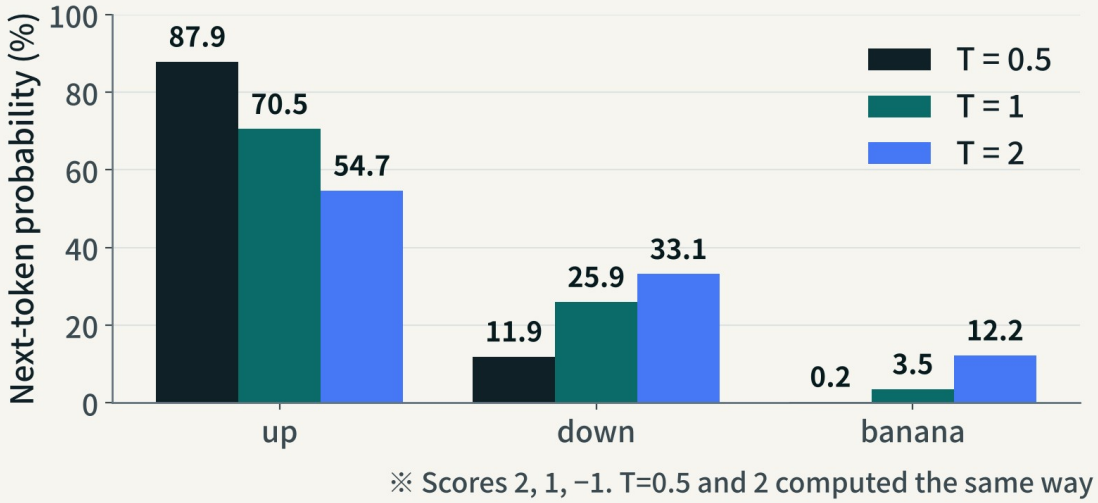
Scores after ‘Yesterday the market ___’: rose 2, fell 1, banana −1

Softmax

$$P_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}, \qquad z = \text{candidate score}, \quad T = \text{temperature}$$

With T = 1

- Rose 70.5%, fell 25.9%, banana 3.5%.
- Lower T favors top scores; higher T spreads choices.



※ Scores 2, 1, −1. T=0.5 and 2 computed the same way

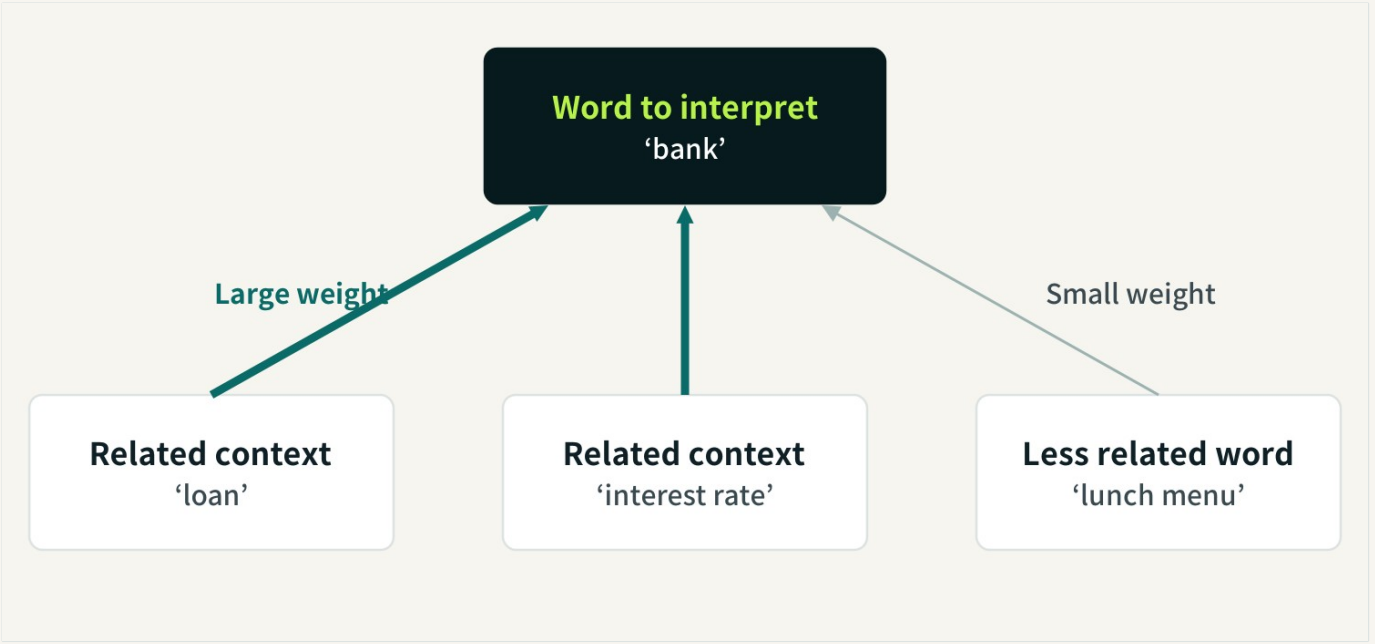
A low temperature does not guarantee factuality

Attention gathers the context it needs, weighted

Interpreting ‘bank’ in ‘The bank changed its loan rate’

Attention

- Each token is a weighted mix of other tokens' information.
- In practice: learned query·key·value vectors, many layers and heads.

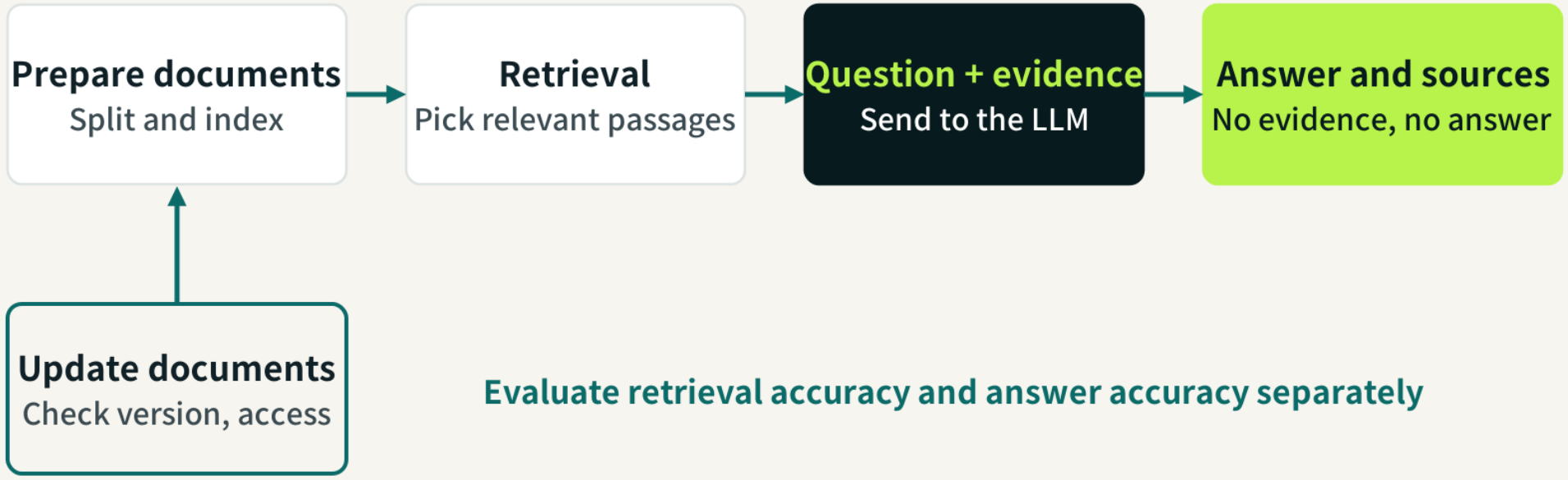


Where in the context to look, and how much

A large weight does not make a token a real cause or a full explanation

RAG answers with relevant documents laid open

Like an open-book exam with the right textbook at hand



Evaluate retrieval accuracy and answer accuracy separately

Link answers to document, page and as-of date

If retrieval has no answer, it says ‘cannot be verified from the given documents’

RAG errs too with wrong documents or missed units

Answering ‘This quarter's op. profit?’ from a quarterly report

Step	Finance example	What to check
Data prep	**Quarterly report**	**Consolidated? Right period and version?**
Retrieval	**‘This quarter's op. profit?’**	**Are related tables and units found?**
Answer	**Explain table figures**	**Mixing up won, millions, 100M?**
Evidence	**Cite page and sentence**	**Does the source back the claim?**

Internal documents do not auto-solve privacy or access issues

Read the retrieval lab apart from full RAG

The source lab (pp.31–32) finds similar documents via TF-IDF and cosine

Source lab = RETRIEVER

Retrieval

- TF-IDF numerically encodes word occurrence.
- Unlike learned semantic embeddings.

Printing matches is a retriever lab

Full RAG

Retrieve + generate

- Complete only when an LLM answers from it.
- The model is not retrained.

Not further training

Even with cited sources, check they back the numbers and claims

03

Lessons from cases

Where finance text and AI actually went wrong

General sentiment lexicons misread financial documents

Loughran and McDonald (2011, Journal of Finance): analysis of US 10-K reports

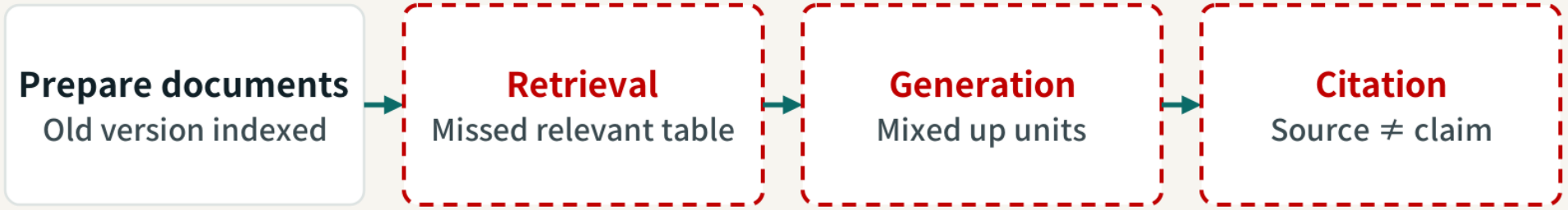
Word	General lexicon	Meaning in finance
liability	**Negative word**	**Neutral balance-sheet term**
tax	**Negative word**	**Routine business item**
cost	**Negative word**	**Basic income item**
capital	**Negative word**	**Neutral capital-structure term**

About 3/4 of general negative words were not negative in 10-Ks

Use lexicons and models validated on financial text

A RAG error arises in one of four steps

Prepare → retrieve → generate → cite: locate the failure to fix it



Check each step separately to find the cause of a wrong answer

On a wrong answer, first check if the retrieved text had it

Right retrieval, wrong answer? Fix generation

Verify even a plausible source in the document

Mata v. Avianca (2023, US): AI-made case citations filed in court

What happened

- A lawyer cited nonexistent cases made by ChatGPT.
- The court imposed sanctions.

In finance

- Match cited filing figures to the page and units.

Design principle

- With no evidence, answer ‘cannot be verified’.

A citation starts the check, not ends it

04

Summary

Close the week with key Q&A

Read text as numbers; verify with evidence

One question, one answer for this week

Check question	Answer
Are close token IDs close in meaning?	**No. Compare meaning with embedding vectors.**
High similarity = correct?	**No. Similarity compares direction, not the odds of being right.**
Does RAG retrain the model?	**Standard RAG only adds retrieved text to the input context.**
A cited source means true?	**Check that the source exists and supports the claim.**

Score retrieval and answers separately

TEXT BECOMES NUMBERS.

ANSWERS NEED SOURCES.

CHECK THE PAGE.

No evidence? ‘Cannot be verified’ is right.